Tree solutions

1

A tree is connected and acyclic. Deleting a leaf inductively from the one-vertex base proves $e=n-1$; equivalently connected graphs need at least $n-1$ edges and forests have at most that many. Conversely an acyclic graph with $e=n-1$ has $c=n-e=1$ by the forest formula, hence is connected and a tree.

2

$L=I+1$ gives $I=30$ and $e=I+L-1=60$. Since $2^4=16<31\le32=2^5$, height is at least five. Take a perfect height-five tree with 32 leaves and contract a deepest sibling-leaf pair to its parent leaf. It remains full, has 31 leaves, and still has height five, proving attainment.

3

The decision tree needs $6!=720$ leaves. Since $2^9=512<720\le1024=2^{10}$, its height is at least ten and $\lceil\log_2 720\rceil=10$.

Split into [5,1,4] and [2,6,3]. On the left split [5] and [1,4]; merging [1],[4] costs one, then compare 5 against 1 and 4 for two more, producing [1,4,5] in three. On the right, [6],[3] costs one; [2] versus [3,6] compares only 2,3 once, giving [2,3,6] in two total. Final comparisons are (1,2),(4,2),(4,3),(4,6),(5,6), then append six. Total is 3+2+5=10.

This is one input's cost. The split has worst-case $M(6)=2M(3)+5=2\cdot3+5=11$. Input [4,2,6,3,1,5] makes both halves use three and final merge five, attaining eleven. Matching the information lower bound once does not prove the algorithm optimal in worst case.

4

Merge 2+3=5, leaving 5(new),5,7,11; then 5+5=10, leaving 7,10,11; then 7+10=17; finally 17+11=28. Merge total is 5+10+17+28=60.

One code is 11:0,7:10,5:110,2:1110,3:1111. Its cost is $11\cdot1+7\cdot2+5\cdot3+2\cdot4+3\cdot4=60$. Mean is 60/28=15/7. Swapping left/right labels preserves these optimal lengths.

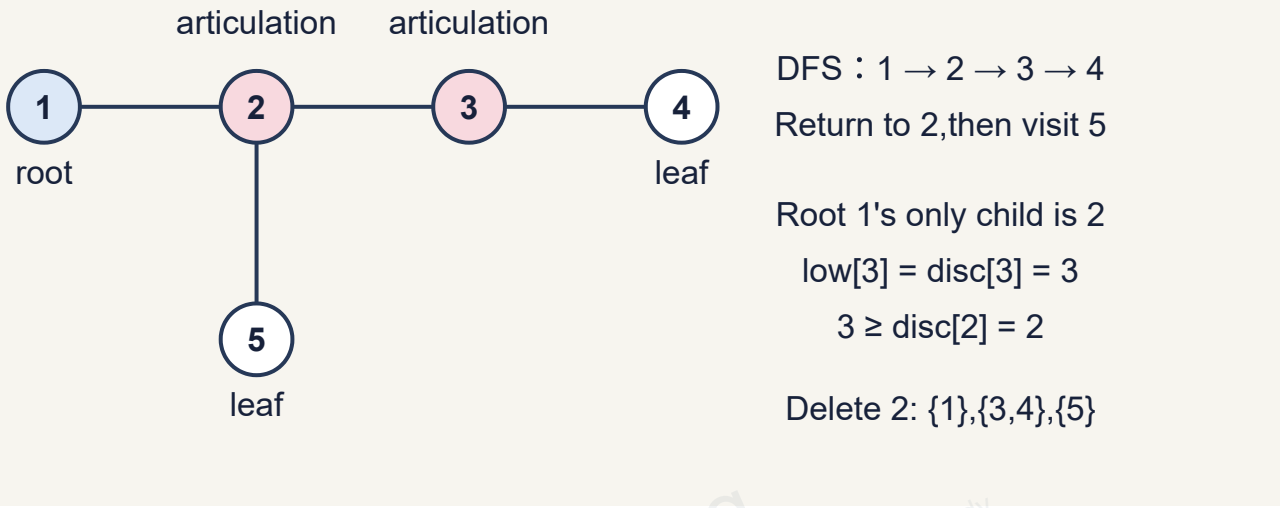
5

Connectivity supplies at least one path. If two distinct simple paths exist, take their first divergence and first later rejoining. The two internally disjoint intermediate paths form a simple cycle, contradicting acyclicity. A shortest nonzero closed walk alone is not enough: traversing one undirected edge out and back gives a length-two closed walk without a simple cycle. Thus there is exactly one simple path.

Tree solutions (continued)

6

Take 1–2–3–4 and add 2–5. With root one, its only child is two. Deleting nonroot two separates vertex one from the remaining branches; child three has low[3]≥disc[2] because no back edge reaches the root. Deleting three also separates four, so cuts are {2,3}. The root's one child does not prevent cuts elsewhere.



7

IDs are e_1=AB, e_2=BC, e_3=CA, e_4=CD, e_5=DE, e_6=EF, e_7=FD, e_8=DG. From A, the discovery order A, B, C, D, E, F, G has disc 1,2,3,4,5,6,7 and parents None, A, B, C, D, E, D.

Push 1,2,3,4,5,6,7. Back edge three makes low[C]=1; back edge seven makes low[F]=4,propagating low[E]=4. Returning E→D satisfies 4≥4: pop 7,6,5 for triangle DEF. Push eight; returning G→D satisfies 7≥4: pop eight for DG. Returning D→C satisfies 4≥3: pop four for CD. Returning C→B has 1<2:no pop. Returning B→A has 1≥1:pop 3,2,1 for ABC.

Final lows are 1,1,1,4,4,4,7. Nonroot C has child D with 4≥3; D has children E, G with lows≥4,so cuts are {C, D}. Strict inequality gives only bridges e_4,e_8. Every edge belongs to one of the four blocks, with triangles intact.

Rooting at D first explores C then A, B; after that E then F; finally G. Root D has three children C, E, G and deletion leaves ABC, EF, G, so it is a cut. C's AB subtree still cannot bypass C to D, so C remains a cut. Discovery times, lows, and pop order may change; cut set, bridges, and four blocks do not.

Tree solutions (continued)

8

Lengths (1,2,2,3) have Kraft sum 1/2+1/4+1/4+1/8=9/8>1,proving impossibility for every prefix code, rather than only failed guesses. Lengths (2,2,3,3,3,3) sum to 2/4+4/8=1 and permit `00,01,100,101,110,111`. First two start with zero and last four with one; within each group words are distinct and equal-length, hence prefix-free.

For one symbol with a separately known count,ε gives mean length zero. If every word must be nonempty, length at least one is necessary, and `0` attains it. Different specifications have different optima.

9

For m=3,l=13, n=ml+1=40,L=(m−1)l+1=27,e=39. The bound $27 \le 3^h$ gives $h \ge 3$. A perfect height-three ternary tree has levels 1,3,9,27,so internal count 1+3+9=13,leaves 27,and edges 3+9+27=39. This explicit tree attains height three.

10

Positive weights make unary internal nodes strictly wasteful: contraction decreases descendant depths. An optimal three-leaf tree is full, necessarily with depths (1,2,2). If weight w gets depth one, the cost is 2(x+y+z)−w. Choose largest z to minimize, giving W=2x+2y+z. Huffman merges x+y then x+y+z, giving the same sum. Words may be z:`0`,x:`10`,y:`11`.

Exchanging depth-two z with depth-one w changes cost by $w - z \le 0$,so keeping z deep cannot improve it. Ties allow several optima. For x=2,y=5,z=9,the merge weights are seven and sixteen, total 23,and mean 23/16=1.4375 bits.

11

If deleting tree edge uv left a u-to-v path, that alternative simple path plus uv would form a cycle. Thus every tree edge is a bridge. Handshake and e=n−1 give $\sum_v (\deg(v) - 2) = 2e - 2n = -2$.

At $n \ge 2$ no degree-zero vertex exists. Leaves contribute −L, degree-two vertices zero, three degree-three vertices contribute three, and one degree-five vertex three. Hence −L+3+3=−2,giving L=8.

Tree solutions (continued)

At n=1 the only vertex has degree zero and contributes −2,so the sum identity still holds. It is not a degree-one leaf; replacing all negative contributions by −L would miss this term. Under rooted no-children terminology the singleton is one leaf, a different convention.

12

Set disc[u]=1 and discover v on e_1 with disc[v]=2. Skip only parent-edge ID e_1. Edge e_2 still reaches ancestor u, so low[v]=min(2,1)=1. Upon return low[v]=disc[u],not strictly greater, and e_1 is not a bridge. Deleting it indeed leaves e_2 connecting u, v.

Skipping every edge to parent u would wrongly skip e_2,leave low[v]=2,and label e_1 a bridge. Correct edge-stack order pushes e_1 then e_2; low[v]≥disc[u] triggers one pop of e_2,e_1 as a two-edge block. Root u has one child and is not an articulation.

13

All six edges lie on triangles, so none is a bridge. Deleting C leaves components {A,B} and {D,E}; C is an articulation. Deleting A or B leaves the other point attached to C, and deleting D or E is symmetric, so there are no other articulation vertices. The two maximal cut-free connected subgraphs are ABC and CDE. Their block-cut tree is B_1–C–B_2, where B_1=ABC and B_2=CDE. No bridge is removed, so bridge deletion leaves the original connected graph, not the two vertex blocks.

After adding BD, deleting C leaves the path A–B–D–E, which is connected. Deleting any other vertex leaves C present: deleting A retains B–C and triangle CDE; deleting B retains A–C and triangle CDE; deleting D retains triangle ABC and E–C; deleting E retains triangle ABC and D–C. Thus every vertex deletion leaves a connected graph, so no articulation remains and the whole graph is one vertex block. BD lies on cycle B–C–D–B, and the original edges remain on their triangles, so there are still no bridges. The new block-cut tree has one block node and no cut nodes.

14

(a) Exchange labels without changing shape

Stage	a word	b word	c word	d word	Depths a,b,c,d	W
Original	`0`	`10`	`110`	`111`	1,2,3,3	75
Swap a,c	`110`	`10`	`0`	`111`	3,2,1,3	61
Swap b,d	`110`	`111`	`0`	`10`	3,3,1,2	54

Tree solutions (continued)

Initially $W = 2 + 10 + 27 + 36 = 75$. First a pays four more and c eighteen less: $(9 - 2)(1 - 3) = -14$, leaving 61. Then b pays five more and d twelve less: $(12 - 5)(2 - 3) = -7$, leaving 54. Words `110`,`111` make a,b siblings at depth three.

(b) Contraction preserves a cost difference

Reduced words are c:`0`, d:`10`, z:`11`, with weights 9,12,7. Originally a,b contribute $2 \cdot 3 + 5 \cdot 3 = 21$; now z contributes $7 \cdot 2 = 14$, exactly seven less. Other terms stay fixed: $W(U) = 9 + 24 + 14 = 47$ and $54 = 47 + 7$.

For reduced weights 7,9,12, merge 7+9=16, then 12+16=28. Thus $W(U^*) = 16 + 28 = 44$, attained by d:`0`, z:`10`, c:`11`. Expanding z gives:

Symbol	Frequency	Optimal word	Depth	Weighted cost
a	2	`100`	3	6
b	5	`101`	3	15
c	9	`11`	2	18
d	12	`0`	1	12
Total	28	—	—	51

The original minimum is $44 + 7 = 51$, with mean $51/28$ bits. Improving from 54 to 51 requires optimizing the reduced tree; the first contracted cost 47 is not its optimum.

(c) Distinguish objectives and prove global optimality

Fixed two-bit coding costs $2(2 + 5 + 9 + 12) = 56$ and has height two. The optimal Huffman tree has height three but cost 51. Four leaves require height at least two, so minimum possible height is attained without minimizing W. The cost-54 tree already has sibling minima yet exceeds 51: that condition alone is also insufficient.

The exchange lemma gives some original optimum with a,b as siblings. Contraction yields $\mathrm{OPT}(2, 5, 9, 12) = \mathrm{OPT}(7, 9, 12) + 7$. A three-leaf full optimum has depths 1,2,2. Weight 12 belongs at depth one, costing $12 + 2 \cdot 7 + 2 \cdot 9 = 44$. Original OPT=51 therefore equals the legal construction above, completing the proof.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study